

## CHAPTER 23

# Sense receptors

The ability to detect stimuli is essential for our survival. By detecting changes in the internal and external environment our bodies are able to maintain homeostasis so that all cells function at an optimum level. In this chapter we will be mainly concerned with the detection of external stimuli.

**Receptors** detect stimuli and as a result generate nerve impulses. In some cases receptors may be widely distributed over the body, such as touch receptors in the skin. In other cases particular receptors are concentrated in a **sense organ**, as are the light receptors in the eye. If the nerve impulses generated by a receptor are conducted to the cerebral cortex we become consciously aware of the stimulus. This awareness is called a **sensation**. Some impulses from receptors may only go as far as the spinal cord or brain stem and result in involuntary responses such as the one illustrated in Figure 22.13.

### Skin receptors

The skin is sensitive to touch, pressure, cold, heat and pain. The stimulus giving rise to each of these sensations is detected by a different type of receptor in the skin.

**Touch receptors** are either free nerve endings surrounding hair follicles, or Meissner's corpuscles. **Free nerve endings** are just bare dendrites. For touch reception they are arranged around the roots of hairs so that when the hair is moved the receptor is stimulated. **Meissner's corpuscles** are dendrites enclosed by connective tissue. They are located fairly close to the skin surface (Fig. 23.1). Touch receptors are more numerous in certain areas of the skin, such as the fingertips, palms of the hands, lips and tongue.

**Pressure receptors** are sensitive to heavy, deep pressure on the skin. This is because the receptors, known as **Pacinian corpuscles**, are deeper down in the skin than those for touch and are surrounded by a comparatively heavy capsule of connective tissue. They are also sensitive to vibration.

**Pain receptors** occur in almost every tissue of the body. Painful stimuli are detected by free nerve endings which, in the skin, are fairly close to the skin surface.

**Heat and cold receptors** occur in the skin and some other parts of the body, such as the tongue. The exact structure of these receptors is uncertain but it is known that there are separate receptors for heat and for cold. Experiments show that there are 'cold spots' on the skin that do not respond to heat stimulation and vice versa.

### Taste receptors

The receptors for taste, or gustation, are located in **taste buds**, most of which are on the tongue. Taste buds occur in groups at the base of little projections on the tongue called **papillae** (sing. papilla). It is the papillae that give the surface of the tongue its rough texture. Taste buds are oval-shaped capsules inside each of which are up to twenty receptors or **gustatory cells**. Each gustatory cell has a hair-like projection, a **gustatory hair**, that projects through a small opening in the capsule (Fig. 23.2).

Only substances dissolved in the saliva in the mouth will cause a taste sensation. The solution enters the taste bud and in some way affects the gustatory hairs. Four taste sensations can be recognised—sweet, sour, bitter and salt. Each of these tastes is detected more strongly by

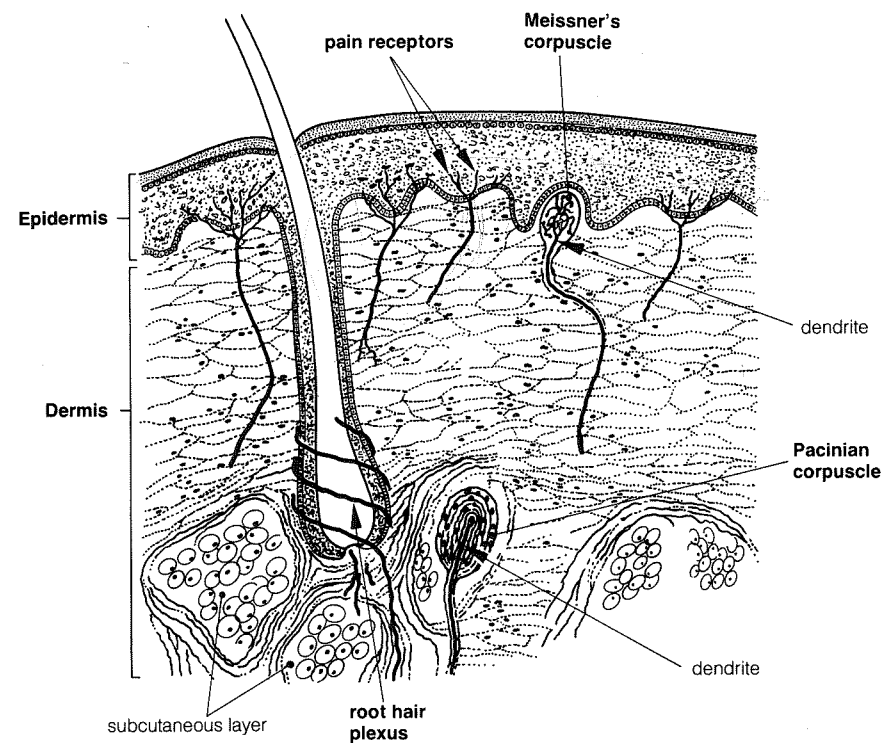


Fig. 23.1 A section through the skin showing the location of some skin receptors

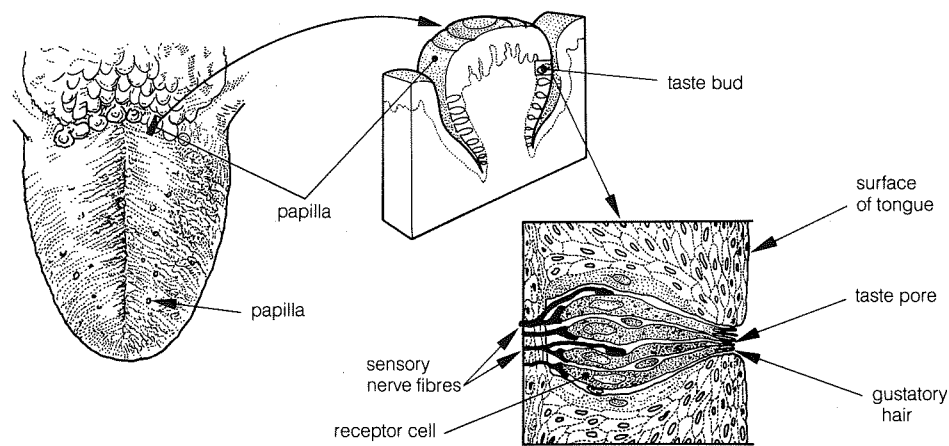


Fig. 23.2(a) Structure of a taste bud. Buds are grouped together in papillae, which occur on the surface of the tongue

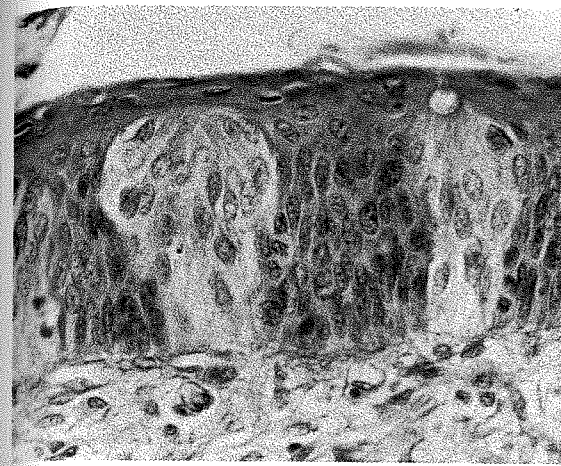


Fig. 23.2(b) Photomicrograph of taste buds  
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OF WESTERN AUSTRALIA

particular areas of the tongue (Fig. 23.3). All the different tastes we can recognise result from either combinations of the four basic tastes or interaction of taste with the sense of smell. There is a close relationship between taste and smell and often what we claim to be taste is actually smell. When a person has a heavy cold and mucus blocks the smell receptors in the nose, taste is often impaired.

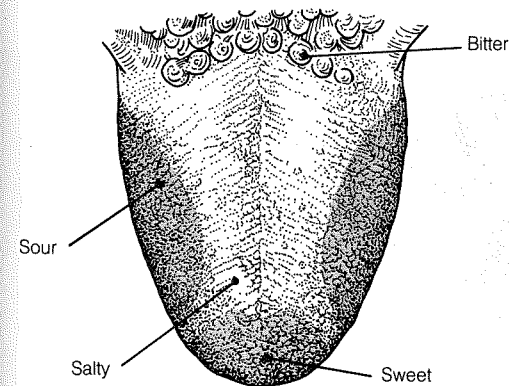


Fig. 23.3 Areas of the tongue most sensitive to each of the four taste sensations

## Smell receptors

Receptors for smell, or olfaction, are specialised neurons that lie in the tissue lining the upper part of the nasal cavity. These neurons have many fine dendrites, or **olfactory hairs**, embedded in the mucous lining of the cavity. There are several million of these receptors, which are not arranged in groups, as are the taste buds, but which are spread fairly evenly. During light breathing air does not pass through the upper part of the nasal cavity. Gases must diffuse upwards to stimulate the olfactory receptors. Sniffing draws air into the olfactory region and enhances the sense of smell (Fig. 23.4).

Although sense of smell in humans is poor compared with other animals we can distinguish thousands of different odours. It would not be possible to have a different receptor for each type of substance, but it is thought that there may be twenty or thirty types of receptor which are stimulated in differing proportions to produce the large number of different smell sensations. Unlike taste, the sense of smell adapts rapidly so that after a few minutes' exposure to a particular odour we can no longer smell it.

## Vision

The receptors for vision are located in the eye. Each eye is well protected by being housed in deep depressions in the skull called the **orbits**. Six muscles, attached to the walls of the orbit, move each eyeball (Fig. 23.5).

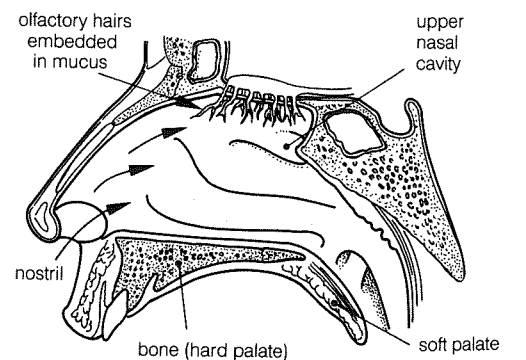


Fig. 23.4 Location of olfactory hairs in the upper part of the nasal cavity

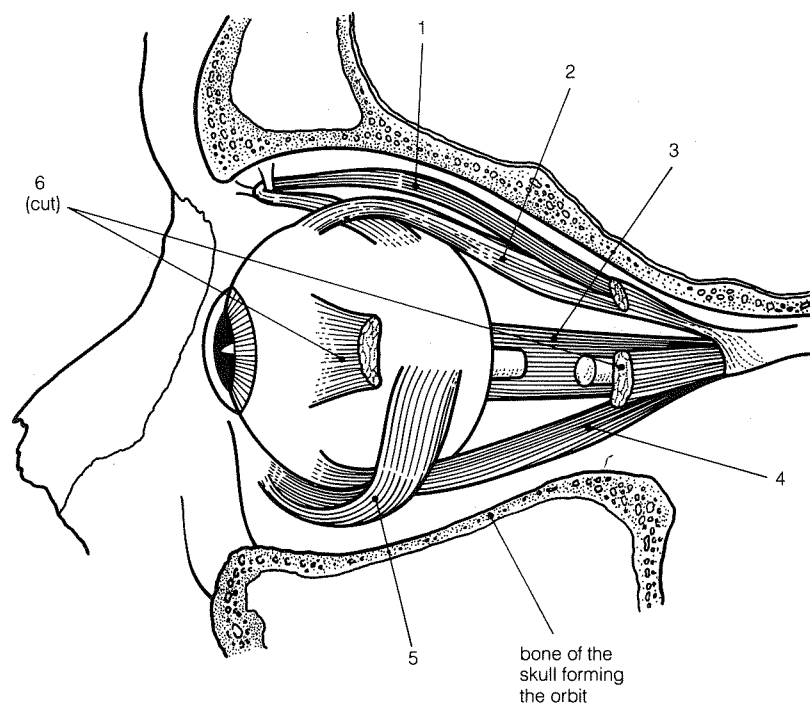


Fig. 23.5 Position of the eye in the orbit, with the six muscles that move the eyeball

### The structure of the eye

The eyeball is composed of three layers of tissue: an outer fibrous layer, the **sclera**; a middle vascular layer, the **choroid** and **iris**; and an inner layer, the **retina**.

#### The sclera

The **sclera** we recognise as the white of the eye. It is tough and fibrous, covering the whole of the eyeball except for the part at the front called the **cornea**. The cornea is transparent and has a greater curvature than the sclera so that it protrudes slightly (Fig. 23.8).

#### The choroid

Inside the sclera is the **choroid**. It is called the vascular layer because it contains many blood vessels. Most of the inside of the sclera is lined by the choroid. It contains a lot of the dark brown pigment, melanin, to absorb light and prevent it being reflected inside the eyeball. At the front of the eyeball, around the edge of the cornea, the choroid forms the **ciliary body**, a structure that contains involuntary **ciliary muscles**. A

continuation of the choroid, called the **iris**, extends beyond the ciliary body. The iris is pigmented and gives the eye its colour. In the centre of the iris is the **pupil**, an opening through which light enters. Muscles in the iris are able to contract and increase or decrease the diameter of the pupil.



Fig. 23.6 The eye from the front, showing the eye and the structures around it  
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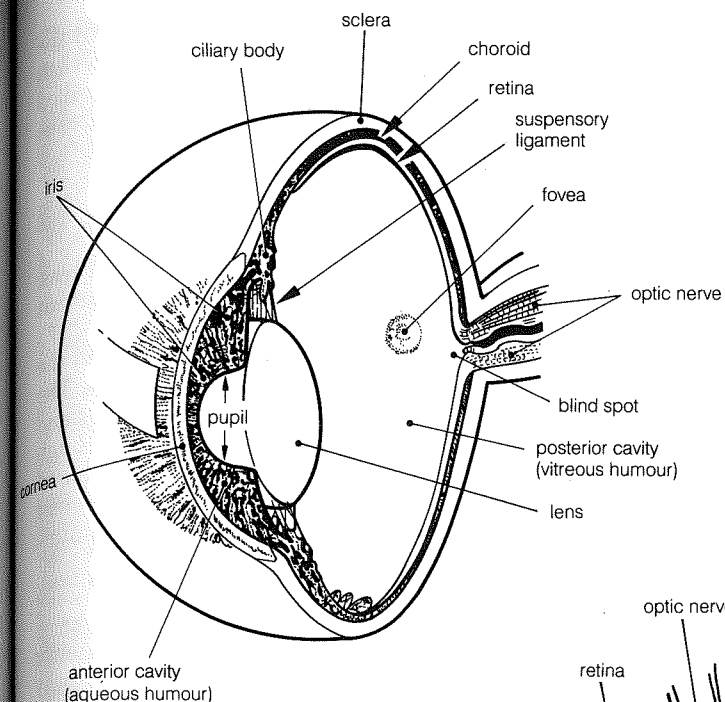


Fig. 23.7 Structure of the eye

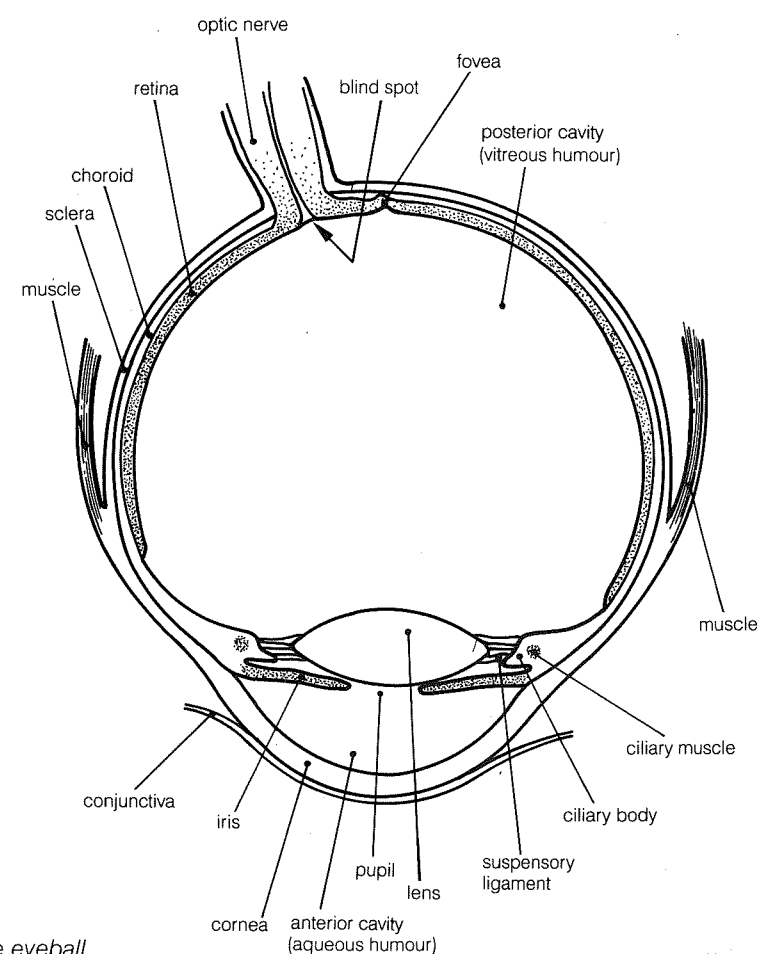


Fig. 23.8 Section of the eyeball



### The retina

The **retina** is the innermost layer of the eye. It lines the rear portion of the inside of the eyeball, extending as far forward as the ciliary body. Within the retina are the nerve cells responsible for vision. The outer parts of these photoreceptor cells are called rods or cones depending upon their shape. **Rods** are cylindrical nerve endings specialised for vision in dim light. **Cones** are more conical in shape (Fig. 23.9). They are only stimulated by bright light and are able to distinguish colours. In the centre of the retina, at the back of the eyeball, is an area known as the **yellow spot**, or macula lutea. A small depression called the **fovea** is in the middle of the yellow spot. Cones are most densely concentrated in the fovea and there are no rods in either the fovea or the yellow spot. Thus the fovea is the area of sharpest vision and when we look directly at an object, light reflected from the object is focused onto the fovea. There are about 7 million cones and 120 million rods in each retina.

The rods and cones actually lie behind the nerve fibres and connector neurons that carry the nerve impulses to the brain. Light entering the eye must pass through these structures before stimulating the photoreceptors (Fig. 23.7). From the rods and cones the nerve impulses are passed to connector neurons and from thence to the dendrites of neurons that will carry the impulses to the brain. The axons of these neurons are

bundled together to form the **optic nerve**. At the point where the optic nerve leaves the eyeball there are no photoreceptors and this area is called the **blind spot**. Any light falling on this area is not detected (Fig. 23.10).

### The lens

Behind the pupil is a biconcave, transparent **lens**. It is held in position by a **suspensory ligament** which joins it to the ciliary body (Fig. 23.7). Contraction and relaxation of the muscles in the ciliary body changes the shape of the lens. The lens is made up of concentric layers of fibres and epithelial cells. Because light must pass through it unhindered, the lens is not able to have a blood supply. The cells in the lens get their nourishment from fluids called aqueous and vitreous humours.

### Cavities and humours

The lens separates the eyeball into an anterior and a posterior cavity. The anterior cavity between the lens and the cornea contains a watery fluid, the **aqueous humour**. This fluid, similar in composition to blood plasma, supplies the cells of the lens and cornea with nutrients and oxygen. The posterior cavity between the lens and the retina is filled with a jelly-like fluid, the **vitreous humour**. Both the aqueous and vitreous humours help to maintain the shape of the eyeball.

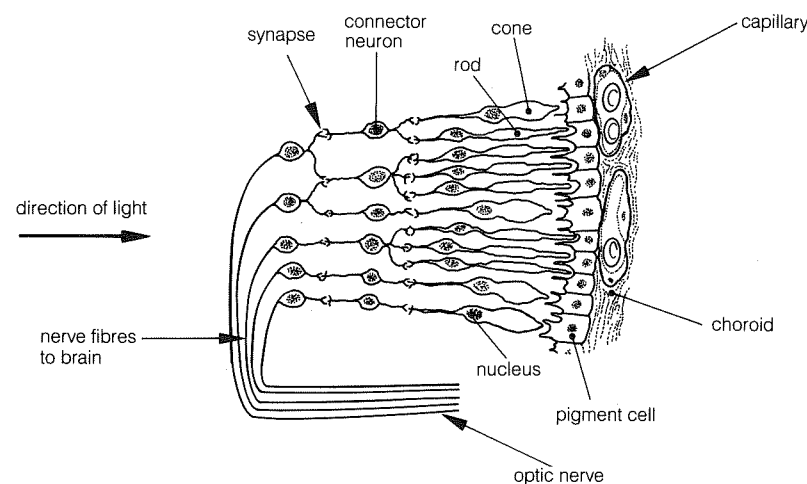


Fig. 23.9(a) Structure of the retina

### Protective structures

As mentioned earlier the eyes are embedded in bony eye sockets and the eyeball itself is formed by the sclera, a very strong, fibrous coat. In addition to these structures the eyelids, conjunctiva and tear glands also play a role in protecting the eyes from damage. The **eyelids** can be closed over the eyes to prevent damage from foreign objects or intense light. Regular blinking keeps the exterior surface of the eye moist and free of dust.

On the inside of the eyelid, and covering the external surface of the cornea, is a thin membrane, the **conjunctiva**. This is a further protective layer for the delicate cornea.

Above each eye is a **tear gland** with ducts that carry the tears to the surface of the conjunctiva. Tears contain salts and an enzyme, lysozyme, which kills bacteria. They keep the surface of the conjunctiva moist, wash away dust and other particles and help to prevent bacterial infection. Tears normally evaporate, but any excess is drained away through a duct that opens at the corner of the eye next to the nose and carries the fluid to the nasal cavity. Thus, when tears are produced in large quantities, for example, when a person cries or when the eye is irritated, it is often necessary to blow the nose.

### Physiology of vision

Vision depends on light rays reaching the retina and stimulating the photoreceptors. To reach the photoreceptors light rays must pass through the conjunctiva and cornea, aqueous humour, pupil, lens, vitreous humour and outer layer of the retina. Furthermore the light rays must be focused to form a sharp image on the retina, just as a slide projector must be focused to produce a sharp image on a screen, or a camera is focused so that the light produces a sharp image on the film.

### Stereoscopic vision

Humans, like most other members of the order Primates, have forward facing eyes. This means that there is considerable overlap between what is seen with the left eye and what is seen with the right eye (Fig. 23.11).

Because the two eyes are a few centimetres apart, each eye sees a slightly different view of

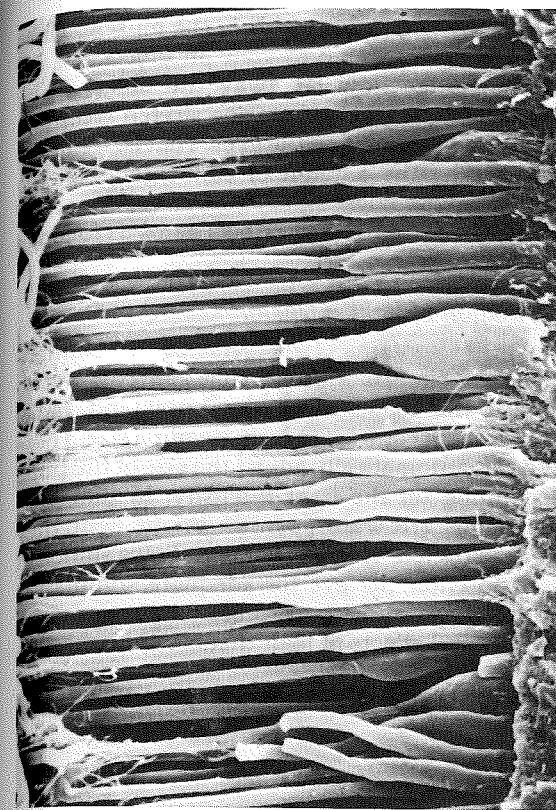


Fig. 23.9(a) Scanning electron micrograph of the retina showing numerous rods and one cone. The pigmented layer is on the right PAUL McMENANIM

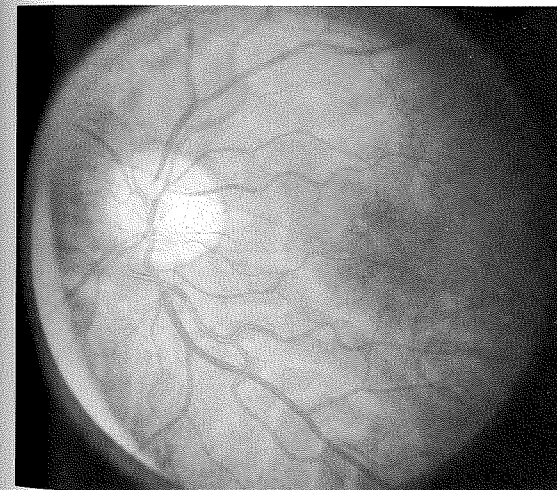


Fig. 23.10 Interior of the eyeball showing retina, blind spot and fovea IAN CONSTABLE, LIONS EYE INSTITUTE

an object. Try holding a pen about 30 centimetres from your face, and looking at it with first one eye, then the other. Notice how the left eye sees slightly more around the left side of the pen than the right eye does, and vice versa. The images formed by each eye are superimposed by the brain and, because each is slightly different, objects appear to have depth as well as height and breadth. That is, we see in three dimensions. This is called **stereoscopic** or **binocular vision**.

To focus on close objects each eye must turn inwards. Have someone hold a pen at arm's length and ask the person to look at it while moving it slowly towards the face. Observe the person's eyes as they turn inwards to keep the pen in view. This is called **convergence**. The nearer the object, the greater the amount of convergence required to maintain binocular vision. Movement of the eyes is brought about by the six muscles, attached at one end to the eyeball and at the other to the bones of the orbit (Fig. 23.5). The muscles contain receptors able to

record the degree of stretch in the muscle. Thus, as the eyes converge to view an object, we are able to judge how far away the object is by interpreting impulses from the stretch receptors in the muscles moving the eyes. That is, we judge distance by the amount of convergence of the eyes. Stereoscopic vision, and our experience of the relative sizes of objects, also helps us to judge distance.

#### Regulation of light intensity

The iris contains two sets of involuntary muscles—circular muscles that can contract to reduce the diameter of the pupil and radial muscles that contract to increase pupil size. These muscles automatically adjust the size of the pupil depending on the light intensity. In dim light the pupils are wide open; in bright light they are constricted. This mechanism allows sufficient light to enter the eye for adequate stimulation of the retina and also protects the retina from the harmful effects of light that is too intense.

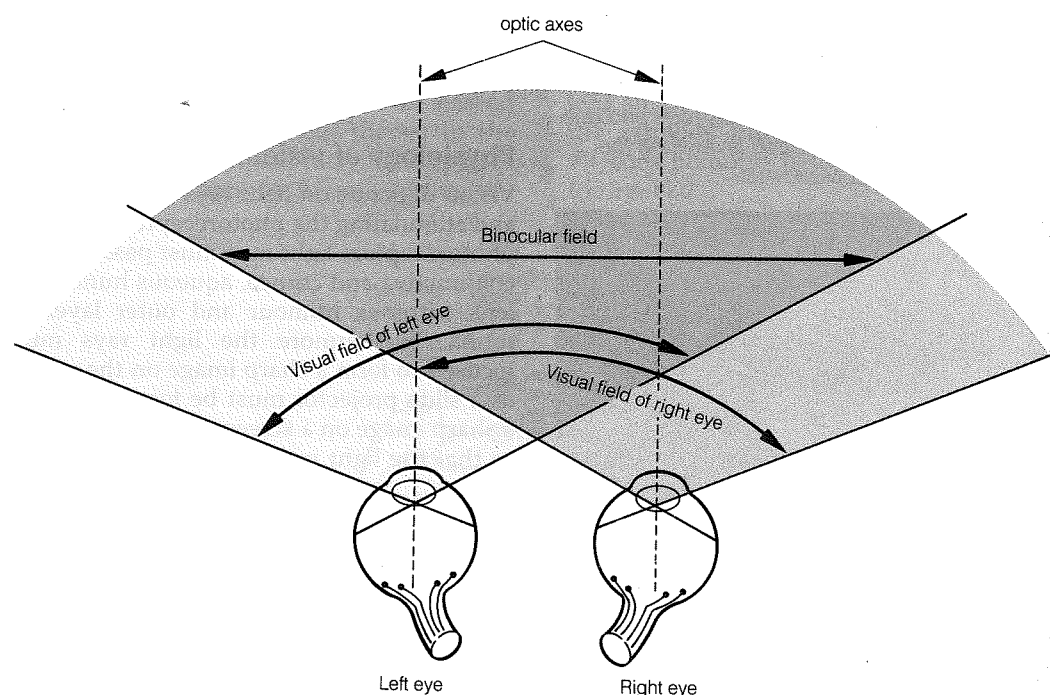


Fig. 23.11 Stereoscopic vision in humans

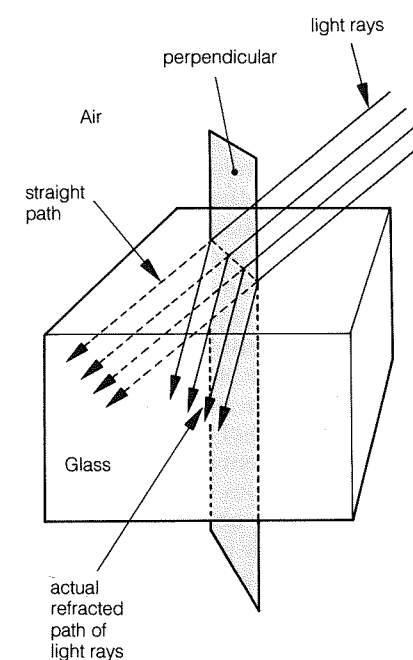


Fig. 23.12 Refraction of light rays as they travel from air to glass

#### Image formation

Light rays normally travel in straight lines. However, if a light ray strikes the boundary

between two media of different density, at an angle other than the perpendicular, the ray is bent. This bending, as light passes from one medium to another, is called **refraction** (Fig. 23.12).

Light passing from the air into the eye is refracted four times before it reaches the retina. Refraction occurs between the air and the cornea, the cornea and the aqueous humour, the aqueous humour and the lens, and the lens and the vitreous humour. Light spreading out from one point on an object can, therefore, be focused to a point on the retina.

As Figure 23.13 shows, the image of an object formed on the retina is upside down. The brain learns, very early in life, to co-ordinate the image formed with the actual position of objects and we do not perceive what we see as being upside down.

Observe Figure 23.13 again. What would happen if the pencil were moved closer to the eye? The light rays diverging from the tip of the pencil would strike the eye at a greater angle and would have to be bent more to focus them to a point on the retina. In a camera the lens always bends the light rays by the same amount, and so the camera is focused by changing the distance between the lens and the film. In the eye the distance between the lens and the retina cannot be changed and so the degree of bending

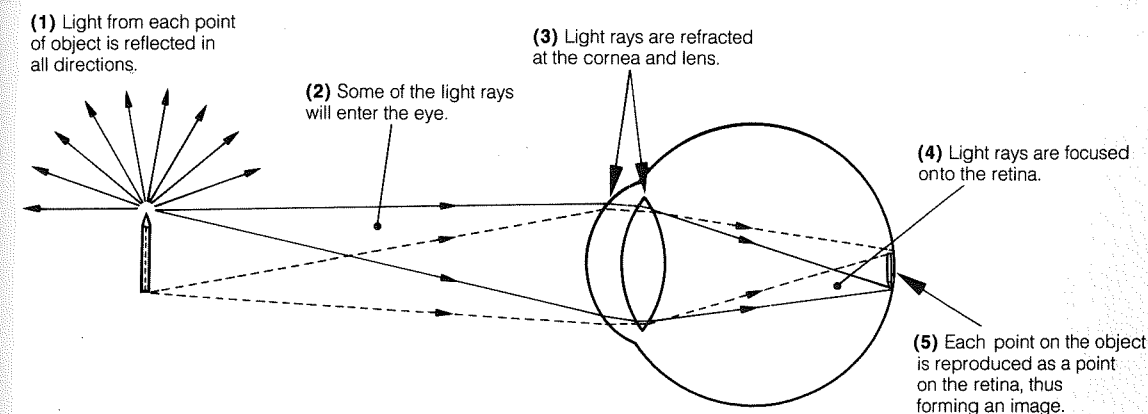


Fig. 23.13 Light rays from each point on an object are focused to a point on the retina. In this diagram only rays from the two ends of the object are shown

of the light rays must be changed. This is done by changing the shape of the lens. The greater the curvature of the lens, the more the light rays are bent towards each other.

Light rays reflected from an object 6 metres or more away from the eye are almost parallel to each other. The lens and other refractive structures bend these rays so that they fall on the fovea where vision is sharpest. If an object is closer than 6 metres the light rays reflected from it will be diverging rather than parallel. To bend these light rays so that they fall on the fovea the lens must become more rounded. This ability of the lens to change shape and focus light rays from objects as close as 25 centimetres, or from objects many metres away, is called **accommodation**. The ciliary muscles are responsible for adjusting the shape of the lens. When a person views close objects, such as when reading or writing, the ciliary muscles are contracted and the lens is bulging. When someone looks into the distance, the ciliary muscles are relaxed and the lens is flatter (Fig. 23.14). As a person gets older the lens gradually loses its elasticity and can no longer accommodate for near and distance vision. The diameter of the pupil also has a role in accommodation. When viewing close objects the pupil is contracted so that light only enters the centre of the lens. This sharpens the image formed on the retina.

#### Stimulation of rods and cones

The stimulus of light rays striking the retina must be converted into a nerve impulse that can be

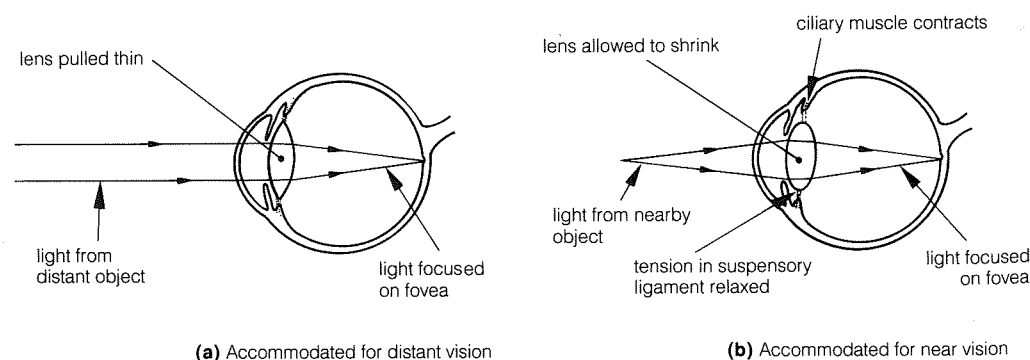


Fig. 23.14 Accommodation of the lens for near and distant viewing

carried to the brain. Rods contain the pigment **rhodopsin**, or **visual purple**, so called because of its reddish-purple colour. Under the influence of light, rhodopsin breaks down into a protein, scotopsin, and retinal, a substance derived from vitamin A. This chemical decomposition results in the formation, in the rods, of nerve impulses that are then conducted away to the brain. Rhodopsin is highly sensitive to light and so the rods enable us to see in very dim light.

Cones contain similar light-sensitive chemicals but they require bright light for their breakdown. There are three types of cones, each with different pigment and each sensitive to a different wavelength (colour) of light—blue, green or red. Stimulation of different proportions of these three types of cones enables us to distinguish colours. Humans share this ability to see colours with other members of the primate order. It is believed that many other mammals can only see black, white and shades of grey.

#### Transmission to the brain

Nerve impulses generated in the photoreceptors are carried by connector neurons to the nerve fibres that make up the optic nerve. Each optic nerve contains about a million fibres and so each fibre must carry impulses from a large number of receptors. (How many rods and cones are there in each eye? See p. 274.) The impulses are taken to the thalamus and from there passed to the visual areas in the cerebral cortex.

## Defects of vision

### Myopia

**Myopia** is short-sightedness. The myopic person can see objects that are close clearly but objects in the distance appear fuzzy. This is because only light rays from close objects can be brought to a focus on the retina. Rays from distant objects are focused in front of the retina. The usual cause of myopia is that the eyeball is too long. Most cases can be corrected by spectacles with concave (diverging) lenses worn for distance viewing. The lenses cause parallel rays to diverge slightly before they enter the eye and they can then be focused accurately onto the retina (Fig. 23.15).

### Hypermetropia

**Hypermetropia** is long-sightedness. The hypermetropic person can see distant objects clearly but has difficulty with close objects. Light rays from distant objects are focused onto the retina but the lens cannot accommodate enough to focus rays from close objects onto the retina. Usually the condition occurs because the eyeball is too short. Spectacles with convex (converging) lenses can be used to correct most cases of hypermetropia. They cause parallel light rays to converge slightly before entering the eye and the

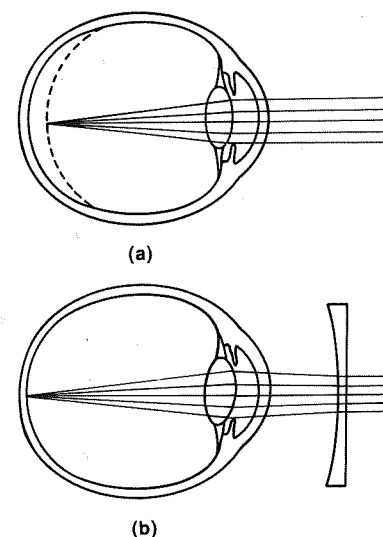


Fig. 23.15 Myopia: (a) parallel light rays are focused in front of the retina and (b) correction by use of spectacles with concave lenses

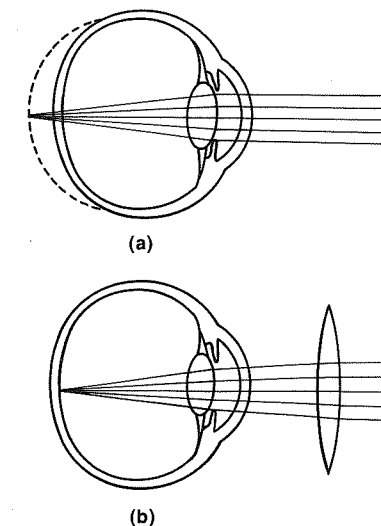


Fig. 23.16 Hypermetropia: (a) parallel light rays are focused behind the retina and (b) correction by means of spectacles with convex lenses

lens of the eye can then converge the light rays to a point on the retina (Fig. 23.16).

### Astigmatism

**Astigmatism** occurs when the curvature of the cornea or the lens is not uniform. For example, the curvature from top to bottom may be different from the curvature from side to side. In such cases it is not possible to see vertical and horizontal lines clearly at the same time. Most cases of

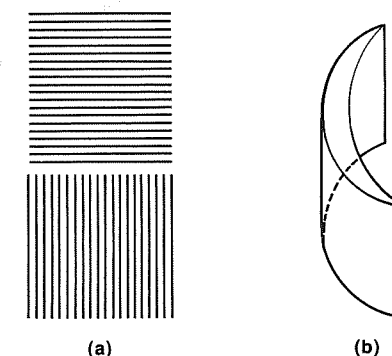


Fig. 23.17 (a) An astigmatic person cannot clearly focus vertical and horizontal lines at the same time; (b) a cylindrical lens used to correct astigmatism



astigmatism can be corrected by spectacles with a cylindrical lens. This is arranged in such a way that it bends the light in one plane only to compensate for the irregularities in the way the lens of the eye bends light rays (Fig. 23.17).

### Presbyopia

**Presbyopia** results from loss of flexibility of the lens as a person becomes older. The lens gradually becomes unable to bulge sufficiently to bend light rays from close objects to focus them onto the retina. The person has to hold books further and further away in order to read them. Eventually

the book may have to be held so far away that the image of the letters is too small to be recognised. At this stage it becomes necessary to wear spectacles with convex lenses for reading and close work. The lenses converge the light rays slightly before they enter the eye and the eye lens can then complete the bending and focus the rays to a point on the retina. Loss of flexibility of the lens begins at a fairly early age but the greatest deterioration occurs after the age of about 40 years. At some time between 40 and 50 years most people begin wearing spectacles for close work.

## Disorders of the eye

- **Conjunctivitis** is an inflammation of the conjunctiva, the membrane that lines the eyelids and covers the cornea. It can be caused by irritants such as dust, smoke or chlorine in swimming pools, but it is most often the result of a bacterial infection, in which case it is highly contagious.
- **Cataract** is the clouding of the lens of the eye so that vision becomes blurred or, in severe cases, no image at all is produced. It is most prevalent in older people. If the loss of vision is serious the lens is surgically removed and replaced, either by a plastic lens inside the eyeball or by spectacles.



Fig. 23.18 A severe case of cataract  
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- **Colour blindness** is an inherited disorder. It is sex-linked (see Chapter 4) and, therefore, occurs in

about 8 per cent of males but only about 0.5 per cent of females. Complete inability to distinguish colours is rare. The most common form of colour blindness is a failure to discriminate between brown, red and green. Colour blindness occurs because one or more of the three types of cones is either absent or does not function correctly.

• **Glaucoma** is another eye disorder that becomes more common with advancing age; 95 per cent of people affected are over 40 years of age. It is caused by an accumulation of fluid in the anterior cavity of the eyeball. This presses the lens into the vitreous humour, which in turn presses on the retina. Prolonged pressure can destroy the neurons and cause blindness. Treatment can be by either drugs or surgery. Drugs are used to cause contraction of the muscles of the iris and ciliary body. This improves drainage from the anterior cavity. Surgery involves removal of a small part of the iris so that drainage is improved. In Australia, the Lions Club Save Sight Foundation holds regular community clinics where people can be tested for glaucoma free of charge.

• **Trachoma** is one of the most common diseases in the world. In Australia, the disease is most common among inland Aborigines where in some communities up to 80 per cent of pre-school children are affected. It is a bacterial infection common in dry, dusty areas. Follicles develop under the eyelids and produce scarring which may eventually result in blindness. It may be forty years after the original infection that blindness occurs. Trachoma can be treated with antibiotics.

## Hearing and balance

The ear contains receptors for sound waves and also receptors that detect head position and movement. Thus, the ear is involved in hearing and in the maintenance of the body's equilibrium or balance.

### The structure of the ear

#### The outer ear

The **outer ear** collects sound waves and directs them inwards towards the rest of the ear. On the outside of the head is the **pinna**, a flap of elastic cartilage covered with skin. The pinna directs sound waves into the **auditory canal**, a tube about 2.5 centimetres long, which ends at the ear drum or **tympanic membrane**. Near the external opening of the auditory canal are hairs, and glands that secrete **cerumen** (ear wax). Both hairs and cerumen help prevent foreign objects from entering the auditory canal. The tympanic

membrane separates the outer ear from the middle ear. It is made of fibrous connective tissue and curves outwards, towards the auditory canal (Fig. 23.19).

#### The middle ear

The **middle ear** is a small chamber hollowed out of one of the bones of the skull, the temporal bone. It is filled with air, which enters through the **Eustachian tube**, a tube connecting the middle ear with the upper part of the throat (nasopharynx). The purpose of the Eustachian tube is to allow air pressure to equalise on both sides of the tympanic membrane. You have probably experienced your ears 'popping' when you drive up or down a high hill. This is the equalising of pressure as air enters or leaves the middle ear through the Eustachian tube. As you go up a hill the external air pressure becomes less and the outward bulge of the ear drum increases because of the higher pressure in the middle ear. Swallowing opens the Eustachian tube so that some of the air in the middle ear

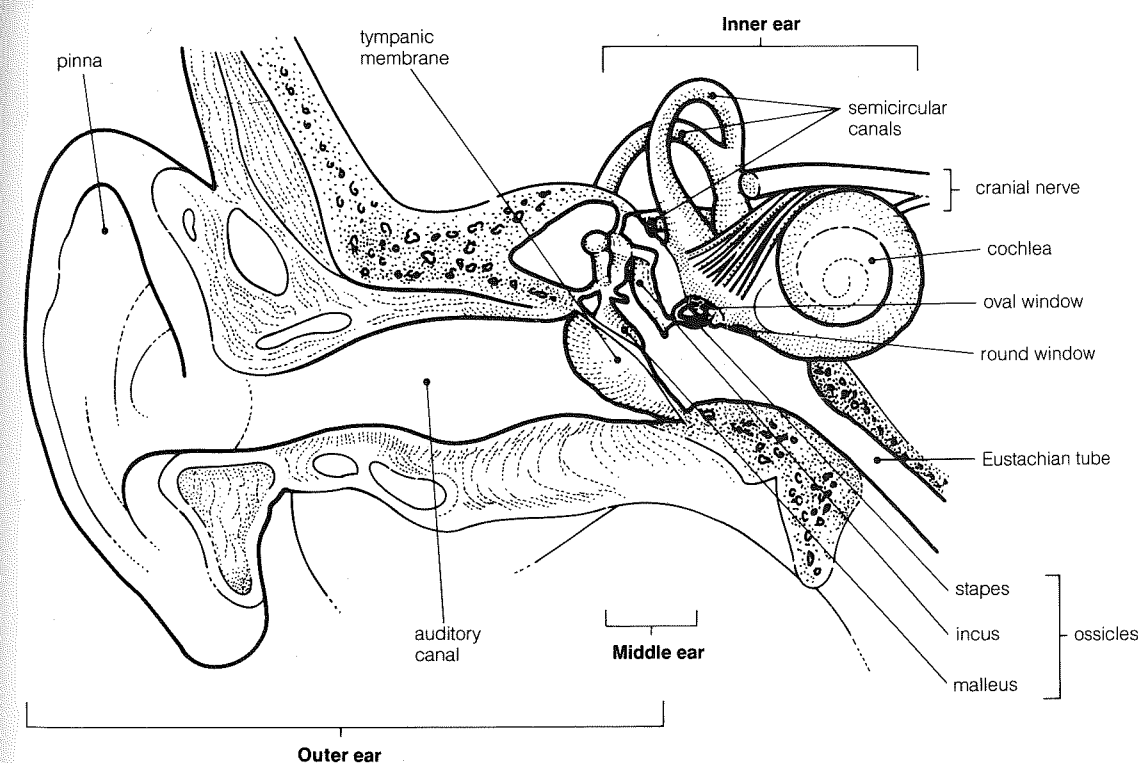


Fig. 23.19 Structure of the ear

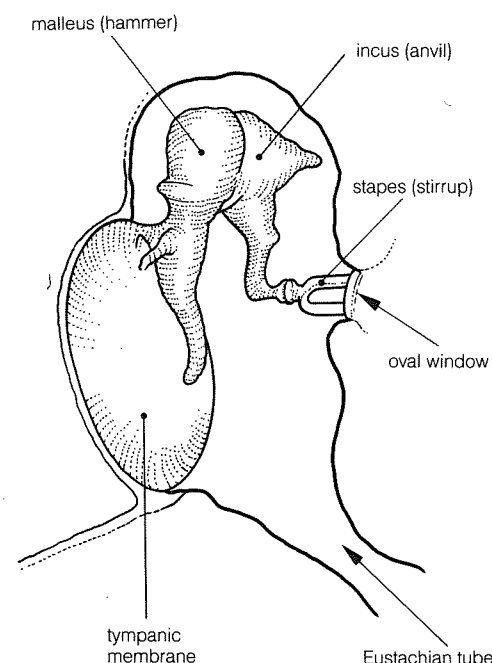


Fig. 23.20 Structures of the middle ear

flows out. This equalises the pressure on both sides of the tympanic membrane and the ears 'pop' as the membrane returns to its normal position. What would happen if you were going downhill?

Inside the middle ear are three tiny bones, the **auditory ossicles**. These are the **malleus**, **incus** and **stapes**, or hammer, anvil and stirrup as they are commonly called because of their shapes. The three bones are joined together to transfer movement from the tympanic membrane, across the middle ear to an opening between the middle and inner ear, the **oval window**. Just below the oval window is another opening, the **round window**. Both of these openings are covered by a membrane that separates the middle and inner ear (Fig. 23.20).

#### The inner ear

The **inner ear** consists of a complicated series of canals and for this reason is called the **labyrinth**. The hollows through the temporal bone that form the canals are the **bony labyrinth**. Inside this and

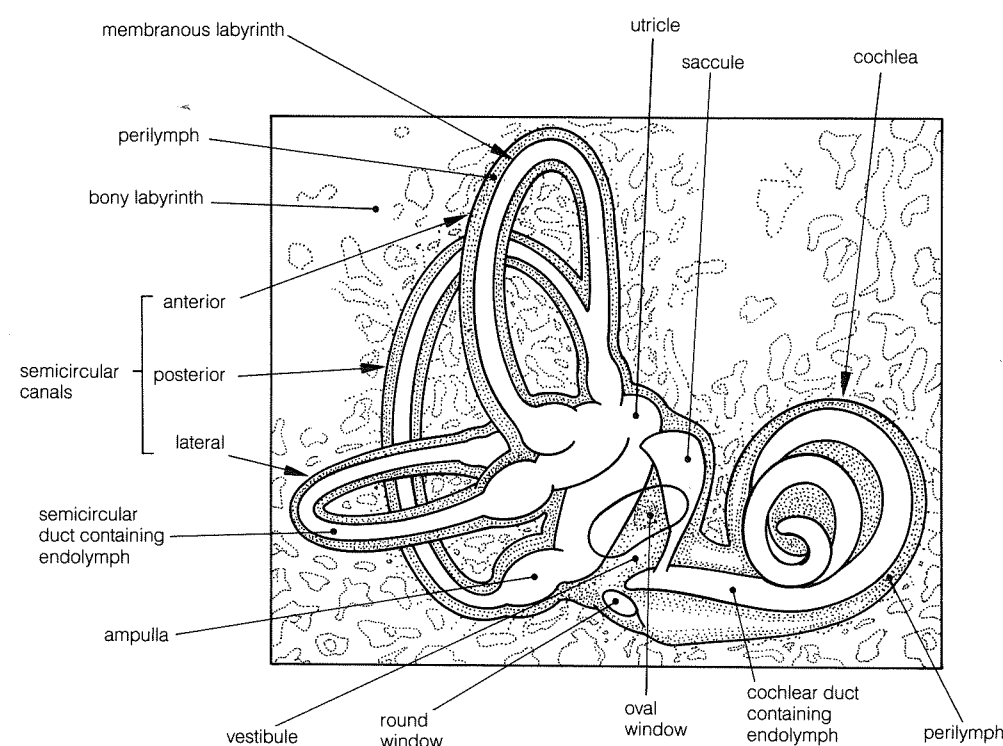


Fig. 23.21 Structures of the inner ear

following the same shape as the bony canals is a **membranous labyrinth**. The membrane is separated from the bone by a fluid called **perilymph** and within the membrane is another fluid, the **endolymph**. There are three parts to the labyrinth—the vestibule, the cochlea and the semicircular canals (Fig. 23.21).

The **vestibule** is joined to the middle ear by the oval window and the round window. Inside the vestibule the membranous labyrinth forms two sacs, the **utricle** and the **sacculus**. The **cochlea** leads away from the vestibule. It is a canal formed into a spiral like a small shell. The spiral makes about two and a half turns around a central core of bone. Within the spiral, membranes divide the cochlea into three passages. The two larger passages are the **scala vestibuli** and the **scala tympani**. The scala vestibuli begins at the oval window and spirals along to the apex of the cochlea where it joins the end of the scala tympani. The scala tympani lies alongside the scala vestibuli and spirals back through the cochlea to end at the round window (Fig. 23.22). These two passages are part of the bony labyrinth and are filled with perilymph. The third passage

in the cochlea, the **cochlear duct**, lies between the two larger ones and, being part of the membranous labyrinth, is filled with endolymph (Figs 23.22 and 23.23).

Within the cochlear duct is the actual organ of hearing, the **organ of Corti**. It consists of a layer of about 30 000 sensory cells, each with a number of hair-like extensions protruding from the free edge of each cell. The hairs are embedded in a layer of jelly-like material (Figs 23.22 and 23.23). Connected to the base of each cell are nerve fibres which join together to form a branch of one of the cranial nerves.

The **semicircular canals** are the third part of the inner ear. From the vestibule these project upwards and backwards. Each canal consists of a looped tube and each is roughly at right angles to the other two (Fig. 23.21). At one end of each canal is a swelling called the **ampulla**. Inside each ampulla is a raised mound of supporting cells on the surface of which are sensory cells. The sensory cells have a tuft of hairs, which protrude into the ampulla and are covered with a small mass of jelly (Fig. 23.24). The semicircular canals are filled with endolymph.

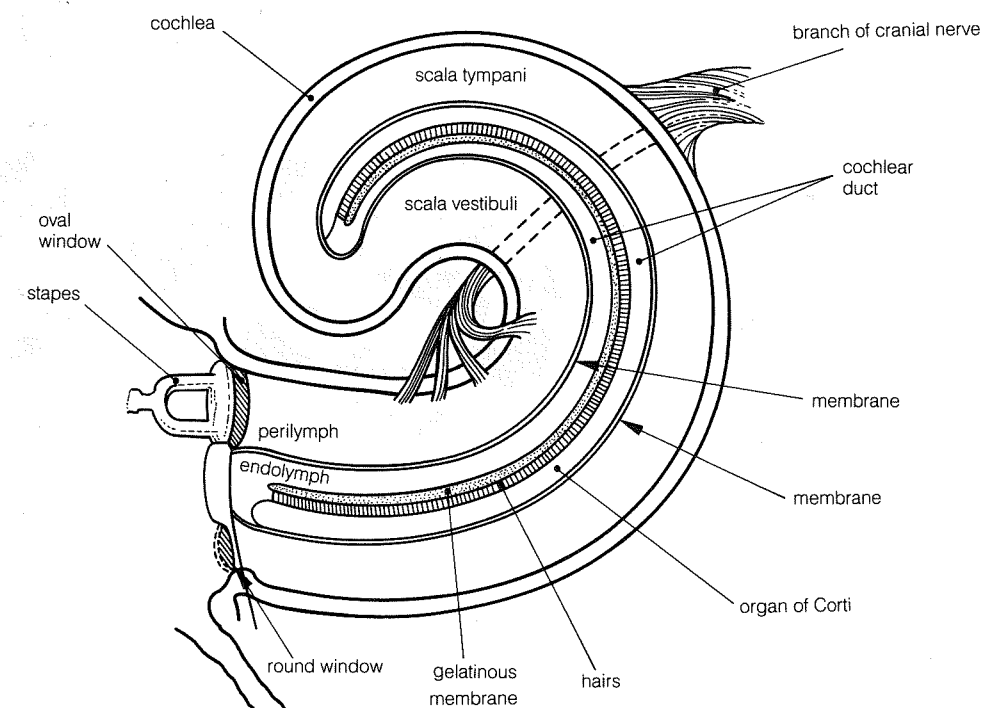
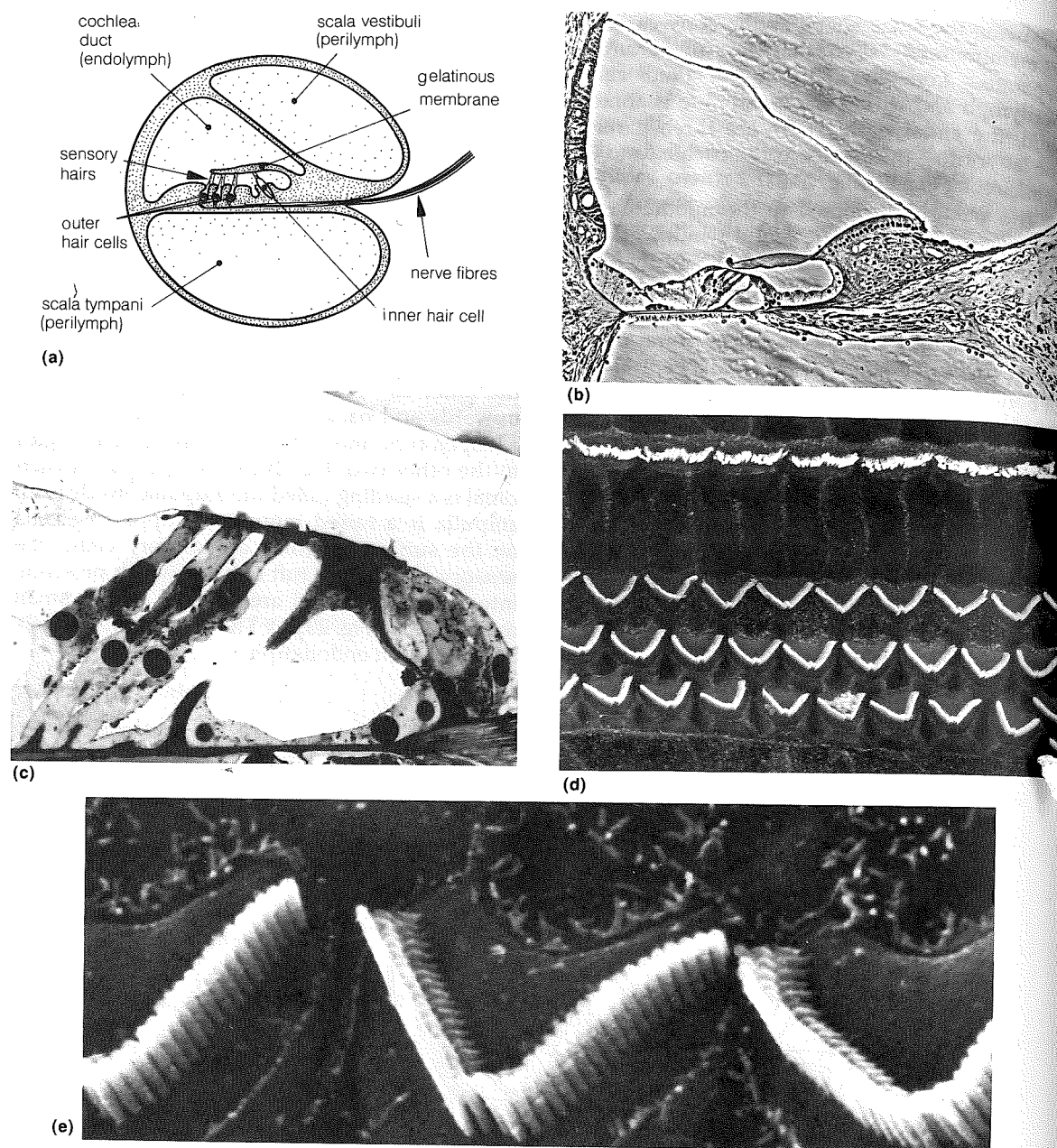
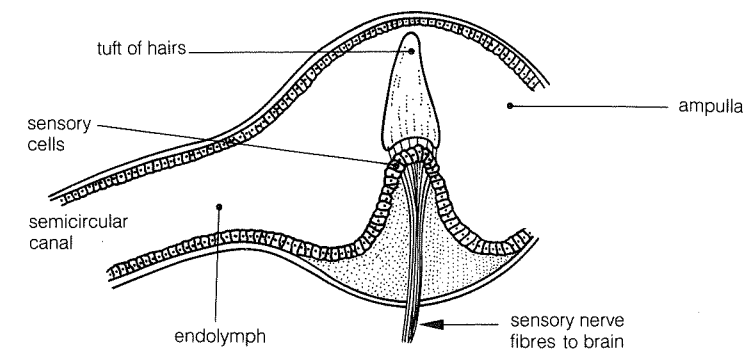


Fig. 23.22 The canals within the cochlea





**Fig. 23.23** Structures of the human body can be examined microscopically at various magnifications and using a variety of microscopic techniques (a) A section through the cochlea showing overall structure (b) Photomicrograph of a section through the cochlear duct and organ of Corti (c) The organ of Corti at higher magnification. Notice the sensory hairs protruding from the surface (d) Scanning electron micrograph looking down on the sensory hair cells. The gelatinous membrane has been removed so that the sensory hairs may be seen (e) Scanning electron micrograph at high magnification showing two sensory hair cells, each with a V-shaped row of sensory hairs. Each hair cell is about 10 micrometres ( $\frac{1}{100}$  millimetre) across PHOTOGRAPHS: AUDITORY RESEARCH LABORATORY, DEPARTMENT OF PHYSIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

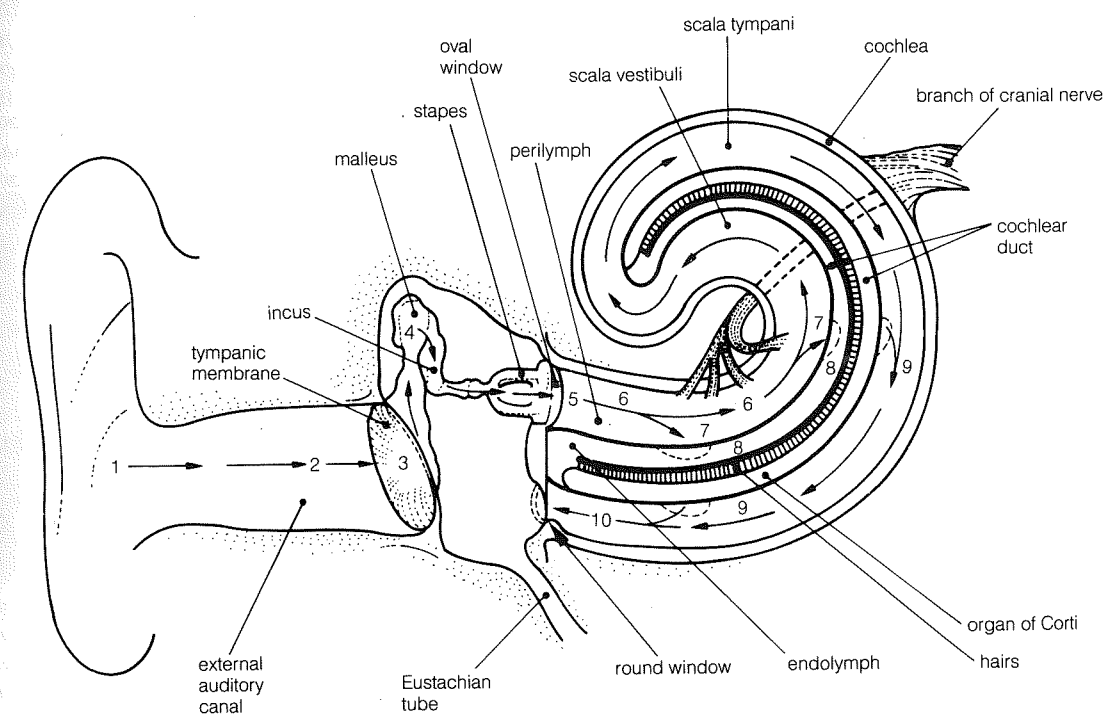


**Fig. 23.24** Section through the ampulla

### Physiology of hearing

Sound is produced by vibrating objects, such as the vibration of a violin string, the vibration of an object when struck with a hammer or the vibration of our vocal cords when we speak. As the object vibrates it produces regions in which the air molecules are squeezed together. These are called **compressions**. Between each compression is a region where the air molecules

are further apart—**rarefactions**. Alternating compressions and rarefactions are the sound waves that spread out in all directions from a vibrating object. The parts of the ear are arranged so that the compressions and rarefactions can be turned back into vibrations and detected by the organ of Corti in the cochlea. Figure 23.25 shows the following sequence of events that occur when we hear a sound.



**Fig. 23.25** Sequence of events in hearing

1. Sound waves reaching the ear are collected by the pinna and directed into the auditory canal.
2. Sound waves travel along the auditory canal to the tympanic membrane.
3. When a compression strikes the tympanic membrane it is pushed in; during the rarefaction that follows, it bulges out again. Thus, the membrane vibrates.
4. Vibration of the tympanic membrane makes the malleus move back and forth. This movement is passed on to the incus and stapes, the three bones acting as levers to transmit the movement across the middle ear.
5. The oval window is pushed in and out by the movement of the stapes.
6. Movement of the oval window causes waves of movement in the perilymph of the scala vestibuli.
7. Movements of the perilymph distort the membranes and bring about movements in the endolymph inside the cochlear duct.
8. Hair cells of the organ of Corti are stimulated, and nerve impulses are transmitted to the brain.
9. At the same time that stimulation of the sensory cells is occurring, movement in the perilymph of the scala vestibuli is transferred to the liquid of the scala tympani.
10. When the movement of the perilymph in the scala tympani reaches the round window, the round window moves in the opposite direction to the oval window. That is, when the oval window moves in, the round window moves out and vice versa. The round window compensates for the movement of the oval window.

Our ears are able to do more than just detect sound. We are able to distinguish sounds of different pitch, we can distinguish soft and loud sounds and we can determine the direction a sound is coming from. Pitch can be distinguished because sound of a particular pitch causes stimulation of sensory cells in one region of the organ of Corti more than in any other region. Volume can be distinguished because loud sounds produce a greater strength of vibration in the cochlear fluids than soft sounds. The stronger the vibration the more hair cells are

stimulated. The direction of the source of a sound can be determined when the sound is to one side of a person's head because the sound will be slightly louder in one ear than the other and the sound reaches one ear slightly before the other.

### Physiology of balance

Receptor cells that give information about the position and movement of the head are located in the saccule, the utricle and in the ampullae of the semicircular canals.

#### Utricle and saccule

Inside the utricle and saccule are sensory hair cells. The hairs are embedded in a jelly-like layer containing small particles of calcium carbonate called **otoliths**. These increase the weight of the jelly so that it presses on the hairs causing the sensory cells to send nerve impulses to the brain. When the head tilts, the otoliths, pulled by gravity, slide downwards. This moves the whole of the layer of jelly, which in turn moves the hairs. A different pattern of nerve impulses is sent from the sensory hair cells to the brain, providing information about the new position of the head (Fig. 23.26).

#### Semicircular canals

The utricle and saccule provide information about the position of the head when it is stationary; the semicircular canals give information about head movements. Because the semicircular canals are arranged in three planes, each at right angles to the other two, they can detect movement of the head in any direction. When the head moves, the semicircular canals move but the fluid inside each canal tends to lag behind. This causes movement of the jelly covering the sensory hairs in the ampulla (Fig. 23.24). The hairs are bent and the nerve impulses initiated are sent to the brain.

### Deafness

Partial or total hearing loss can be classified as either conductive deafness or nerve deafness. **Conductive deafness** is caused by some fault in the conduction of the sound through the outer or middle ear. Common causes are blockage of the auditory canal by accumulated wax or by

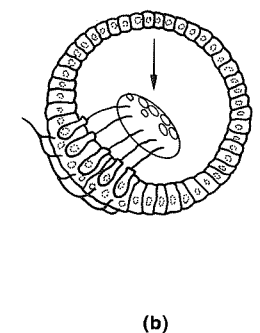
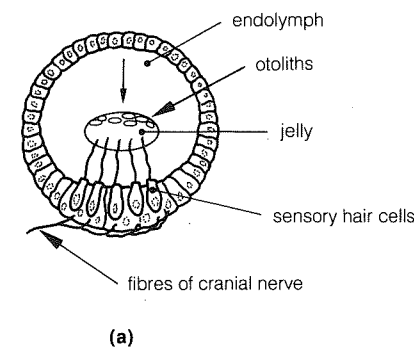


Fig. 23.26 Stimulation of sensory hair cells in the utricle (a) when the head is erect and (b) when the head is tilted

foreign objects, infection causing accumulation of fluid in the middle ear, or the fusing of the ossicles so that vibration is not transmitted from the eardrum to the oval window. Hearing aids that transmit sound to the inner ear through the bone of the skull rather than through the middle ear can be used to overcome some forms of conductive deafness.

**Nerve deafness** is loss of hearing resulting from defects in, or damage to, the sound receptors in the cochlea, the nerve from the cochlea or nerve pathways in the brain. Such hearing loss could be caused by infections, tumours or prolonged exposure to loud noise. Hearing aids may be of little help in such cases. Continual exposure to

loud noises can cause permanent damage to the sensory hair cells (Fig. 23.27). The major source of such noise is workplaces where noisy machinery is in continual use. Wearing of ear muffs is recommended in such situations.

**Gradual hearing loss with ageing**, particularly for higher pitched sounds, is fairly common. It is normally due to a combination of causes, such as reduced elasticity of membranes in the cochlea, reduced blood flow to the cochlea causing poor nutrient supply and gradual degeneration of sensory cells. Hearing aids that amplify sounds can be used to correct such hearing loss.

Some forms of deafness are inherited (see Chapter 5) so that people may be born deaf. For them, as for people who suffer from some forms of hearing loss, hearing aids are of no use. A major advance in medical technology has been the development of the **bionic ear** which can help such people. It consists of a microphone, a processor that converts signals from the microphone into electrical impulses, and electrodes implanted in the nerve from the cochlea. Such devices cannot yet provide normal hearing, but for the profoundly deaf any hearing at all is a major advance.

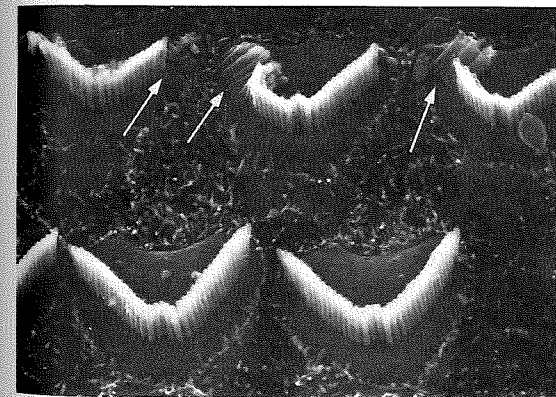


Fig. 23.27 The sensory hairs, indicated by the arrows, have been flattened by prolonged exposure to loud noise AUDITORY RESEARCH LABORATORY, DEPARTMENT OF PHYSIOLOGY, UNIVERSITY OF WESTERN AUSTRALIA

#### Other disorders of the ear

- **Motion sickness**, such as sea, air or car sickness, is caused by excessive stimulation of the sensory cells in the utricle, saccule and ampullae. Repeated movement in a particular direction, such as the up and down or side to side motion experienced at sea, can result in nausea or vomiting. Psychological factors, for example, anxiety or fear, also contribute to motion sickness.



- **Tinnitus** is a noise in the ear. It may come and go, but in severe cases it is continuous. Sometimes the cause, such as inflammation and accumulation of fluid, can be identified and treated. Often there is no apparent cause and no cure.
- **Swimmer's ear** is a bacterial or fungal infection of the outer ear. It can be very painful or maddeningly itchy. Antibiotics are used to kill the infecting micro-organism.
- **Otitis media** or **middle ear infection** is most common in young children. Infecting micro-organisms pass up the Eustachian tube and cause an infection in the middle ear. Young children are more prone to this type of infection because a child's Eustachian tube is straighter than an adult's and is open most of the time. Such infections can be extremely painful and early treatment is recommended because the infection can cause permanent hearing loss or may spread to the brain and cause meningitis. Infections of the outer and middle ear are very common in children. One-third of sick children who visit a general practitioner have an ear infection. A survey carried out in Adelaide showed that eighty-seven per cent of all children had suffered an ear infection by the age of 4 years.

### Review questions

- Describe the differences between touch, pressure, pain, heat and cold receptors.
- (a) Draw a diagram of the tongue showing the location of the areas most sensitive to each of the four types of taste.  
(b) Why is taste sometimes impaired if a person has a heavy cold?
- Where would you find olfactory hairs? Describe their role in the sense of smell.
- Describe the three layers of tissue that make up the eyeball. Explain the function of each.
- Explain the differences between the rods and cones in terms of their shape, distribution, relative numbers and sensitivity to light and radiation.
- The eye is a very delicate organ. Describe the role of protective structures that help prevent damage to the eye.
- Explain the importance of stereoscopic vision and convergence in judging depth and distance.

- (a) Draw, and label, a diagram showing how light from a distant object is focused onto the retina.  
(b) Draw, and label, a second diagram showing how the lens accommodates when the object is brought closer to the eye.
- How is the amount of light entering the eyeball regulated?
- Explain the cause of each of the following defects of vision and describe how each may be corrected:
  - hypermetropia
  - myopia
  - presbyopia
  - astigmatism
- Explain the role of each of the following structures in the ear:
  - tympanic membrane
  - auditory ossicles
  - oval and round windows
  - perilymph and endolymph
  - organ of Corti
- (a) What structures in the ear are responsible for detecting the position of the head? Explain how they work.  
(b) What structures in the ear are responsible for detecting movement of the head? Explain how they work.
- Explain the difference between conductive and nerve deafness and describe some causes of each.

### Apply your knowledge

- Rhodopsin, the light sensitive pigment in the rods of the retina of the eye, is formed from two chemicals, one of which is vitamin A. What are some of the symptoms you would expect a person to have if suffering from vitamin A deficiency?
- To see an object more clearly in the dark it is recommended that you do not look directly at the object, but slightly to one side of it. Can you explain why this would enable you to see the object more clearly?
- Turn around rapidly on the spot several times and then stop. Notice that you have the

sensation that you are still moving. Explain why this sensation occurs.

- (a) Describe three situations in which detection of external stimuli is essential for survival.  
(b) Describe three situations in which detection of stimuli is essential for homeostasis.
- Flying in a light aircraft (which does not have a pressurised cabin) when suffering from a heavy cold could cause damage to the ear. Explain how this could occur.
- The ear functions as both an organ of balance and an organ of hearing. Which of these two functions do you consider to be more important? Give reasons for your answer.
- The pitch of a sound depends on the frequency of the sound wave. A human ear can detect sound waves with frequencies between 20 and 20 000 cycles per second (hertz). Explain how we can easily distinguish the difference in pitch between middle C (264 cycles per second) and D (297 cycles per second) on a piano.
- Does the presence of hair on the skin increase or decrease the skin's sensitivity to touch? Explain.
- If you spilled a hot cup of tea or coffee over your hand it would cause a strong burning sensation, yet you can drink hot tea or coffee without discomfort. Explain why this is so.
- The sense of smell fatigues rapidly but the sense of taste does not. That is, after exposure to a particular odour for a short time we can no longer smell it, whereas we can go on tasting a substance no matter how long the exposure. Can you see any advantages in this difference between taste and smell? Would natural selection have favoured the development of the two senses in this way?
- Refer to Figure 23.6 on page 272.
  - List the protective structures of the eye that you can see in the photograph.
  - Do you think the photograph was taken in bright or dim light? Explain.
- Suggest why it is that some receptors are concentrated in one place (e.g. light, smell and taste receptors) while others are distributed all over the body surface (e.g. touch and pressure receptors).